Fundamentals in AI & ML

Project Report

TOPIC:- AI VIVA Examiner System

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PROJECT REPORT

AI VIVA EXAMINER SYSTEM

- **INTRODUCTION:**

  In today's education system, oral examinations(viva) play a crucial role in evaluating a student's conceptual understanding, communication skills, and confidence. However, traditional viva exams require human examiners, which can lead to inconsistencies , bias and limited availability.

  The **AI VIVA EXAMINER SYSTEM** is designed to simulate a real-time viva examination environment using Pyhton and Artificial Intelligence concepts. The system asks subject-based questions, captures voice responses, evaluates answers, and provides performance feedback instantly.

  This project aims to make viva preparation smart, interactive, and automated helping students practice effectively without needing a human examiner.

- **PROBLEM STATEMENT:**

  Students face several challenges in viva preparation:

  1. **Lack of Practice Environment:**
     Students do not get enough real-time viva experience.
  2. **Fear & Nervousness:**
     Lack of confidence while speaking answers.
  3. **No instant feedback:**
     Students cannot evaluate their answers immediately.

4. **Limited Access to teachers:**
   Not always possible to conduct frequent viva sessions.

➢ This project solves these problems by creating an AI-based automated viva system.

- **<u>OBJECTIVES:</u>**
  I. To simulate a real viva examination environment.
  II. To enable **voice-based answer** input.
  III. To provide **AI-generated structured answers**.
  IV. To give **instant feedback and scoring**.

- **<u>FUNCTIONAL REQUIREMENTS:</u>**
  The system provides the following functionalities:
  I. **User Input Handling:** Accepts class (10th/12th **)** and subject selection.
  II. **Question Generation:** Generates random conceptual questions from the syllabus topics.
  III. **Voice Recognition:** Captures spoken answers using microphone.
  IV. **AI Answer Generation:** Fetches structures answers using Wikipedia API.
  V. **Performance Evaluation:** Calculates accuracy based on keywords matching.
  VI. **Scoring System:** Provides marks out of 10 with performance level.
  VII. **Report Generation:** Displays subject, topic, accuracy and reference answer.

- **NON-FUNCTIONAL REQUIREMENTS:**

  I. **Usability:**
     - Simple CLI-based interface.
     - Easy to use for students.

  II. **Efficiency:**
     - Fast execution using lightweight libraries.

  III. **Reliability:**
     - Handles speech errors and invalid inputs.

  IV. **Scalability:**
     - Easy to add more subjects or topics.

- **SYSTEM ARCHITECTURE:**

  The system follows a modular procedural architecture:

  - **CORE MODULES:**
    1) **Input Module:** Handles user selection and voice input.
    2) **Processing Module:** Topic selection, Question generation and Answer fetching.
    3) **Evaluation Module:** keyword extraction, Accuracy calculation, Score generation.
    4) **Output Module:** Display performance report.

- **DESIGN, DECISIONS AND RATIONALE:**

  1. **Language Selection:** Python is used due to: Simplicity and readability, strong libraries like speech_recognition, Wikipedia, threading.
  2. **Use of Dictionaries:** Stores syllabus topics efficiently and enables dynamic subject expansion.

3. **Algorithm for evaluation:**

Accuracy is calculated as:

Accuracy = (Matched keywords/ Total keywords) X 100

Final score= (Accuracy X 0.07) + (Word count X 0.05)

- **<u>IMPLEMENTATION DETAILS:</u>**
  - **Libraries Used:**
    1. **Time:** delay and timer.
    2. **Sys:** output control.
    3. **Random:** random question selection.
    4. **Threading:** animated listening waves.
    5. **Math:** wave animation logic.
    6. **Speech_recognition:** voice input
    7. **re:** keyword extraction.
    8. **Wikipedia:** fetching answers.

- **<u>KEY FUNCTIONS:</u>**

1. **Welcome ():** Displays instructions.
2. **Voice ():** Captures speech input.
3. **get_random_topic ():** Select topic.
4. **answer ():** Generates structured answer.
5. **check_accuracy ():** Evaluates answer.
6. **analyze_answer_pro ():** calculates scores.
7. **generate_question_and_answer ():** Generates Q & A.
8. **guide ():** Controls full workflow.

- **WORKING FLOW:**

1. User starts program.
2. Selects class and subject.
3. System generates question.
4. 30 seconds preparation timer.
5. User speaks answer.
6. AI analyses response.
7. System displays: Accuracy, Score, Performance level & Reference level.

- **SCREENSHOTS / OUTPUT:**

```
PS C:\Users\Dhruv\OneDrive\Desktop\ai> & "C:/Program Files/Python311/python.exe" c:/Users/Dhruv/OneDrive/Desktop/ai/main.py
============================================================
                      AI VIVA EXAMINER
============================================================

System: Ready for Examination.
Instructions: Answer clearly and technically for better marks. Best of Luck!

Enter Class (10th/12th): 12th
```

```
Enter Class (10th/12th): 12th

Select Subject:
1. Physics
2. Chemistry
3. Biology
4. English
Enter number: 1
```

```
[System] Generating Physics question for viva examination.......
QUESTION: Differentiate Atoms with related concepts in Physics.

Preparation: 30s
Timer: 30s
```

```
SPEAK NOW...
[Listening]
```

```
[System] Generating Physics question for viva examination.......
QUESTION: Can you elaborate on Atoms with examples?
```

```
Your Response: with example

--- PERFORMANCE REPORT ---
Subject: Physics | Topic: Atoms
Accuracy: 0.0% | Grade: Needs Work
Final Score: 0.1/10
----------------------------------

Reference Answer:

DETAILED ANALYSIS: ATOMS

Definition:
Atomic physics is the field of physics that studies atoms as an isolated system of electrons and an atomic nucleus

Explanation:
Atomic physics typically refers to the study of atomic structure and the interaction between atoms. It is primarily concerned with the way in which electrons are a
rranged around the nucleus and
the processes by which these arrangements change. This comprises ions, neutral atoms and, unless otherwise stated, it can be assumed that the term atom includes io
ns.
The term atomic physics can be associated with nuclear power and nuclear weapons, due to the synonymous use of atomic and nuclear in standard English.

Key Points:
- processes
- Atomic
- typically
- concerned
- power

Application:
Used in practical and theoretical understanding of Physics.

[Source: Wikipedia]
```

- **<u>TESTING APPROACH:</u>**

  1. **Unit Testing:** Invalid subject input and No speech detected→ error handled.

2. **Boundary Testing:** Time limit for speech→ checked.
3. **Exception Handling:** Wikipedia errors and microphone failures are handled.

- **SOLUTIONS:**

1. Used try-except for error handling.
2. Added fallback answers.
3. Used regex for better keyword extraction.

- **LEARNING AND KEY TAKEAWAYS:**

1) Practical use of **AI in education**.
2) Learned **speech processing basics**.
3) Improved understanding of **Multithreading**, **NLP basics** and **Algorithm design**.

- **FUTURE ENHANCEMENTS:**

1. GUI using Tkinter or web app.
2. Integration with ChatGPT API (better answers).
3. Database for storing student performance.
4. Real- time analytics dashboard.
5. Multi-language support.

- **CONCLUSION:**

The AI Viva Examiner System successfully simulates a real viva successfully simulates a real viva environment by integrating voice recognition, AI-based answer generation, and performance evaluation.
This project enhances learning by making viva preparation:

Interactive, smart and self-assessable. It demonstrates how AI can transform traditional education systems.

- **REFERENCES:**

1. Wikipedia API.
2. Pyhton Documentation.
3. Speech Recognition library Docs.
4. Academics syllabus of CBSE.